

Jay Gurnani 古杰

Staff Software Engineer / Team Lead

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☰ Skills

Programming

Golang | Java | Python | C# | Typescript | ReactJS

Architecture

Distributed Systems | Event-Driven Systems | GraphQL | gRPC | CI / CD

AI

MCP Servers | LLM Integration | AI Agents | Prompt Engineering

Cloud and Data

AWS | Kafka | Docker | Terraform | DynamoDB | OpenSearch | SQL

Engineering Leadership

Requirements analysis | Sprint planning | People and stakeholder management

🎓 Education

UNSW

MASTER (WITH EXCELLENCE) OF INFORMATION TECHNOLOGY
Nov 2016 | Sydney, Australia
High Distinction Average (Top 9% in class)

BS IN COMPUTER SCIENCE

Nov 2012 | Sydney, Australia

AISG

HIGH SCHOOL - IB DIPLOMA
June 2009 | Guangzhou, China

🌐 Languages

Native proficiency:

English • Mandarin Chinese

Business proficiency:

Hindi

👤 Summary

Engineering team lead with 13 years of professional software development and 2+ years of engineering management experience. Proven track record of scaling products from zero to thousands of daily active users and leading teams delivering platforms handling billions in annual revenue.

📁 Experience

Motional Staff Engineer Team Lead

📅 Dec 2025 –Present 📍 Singapore

Engineering Enablement Team lead tasked with improving developer productivity across the company

Tech Stack: Python, Bazel, C++

- Influenced engineering priorities across multiple teams through **data-driven recommendations and technical leadership**
- Identified and eliminated systemic bottlenecks in build, test and deployment workflows, reducing engineering merge cycle time by **72%**
- Led adoption of internal developer tooling (Backstage) used across multiple engineering organizations, **reducing time-to-production** for new services
- Established company-wide standards for on-call ownership, onboarding **100+** engineers onto a unified operational model
- Defined company initiatives to migrate Jira workflows into new custom built system saving **1 week** in build to release time

TikTok Senior Engineer

📅 Sept 2024 –Nov 2025 📍 Sydney, Australia

Developer for the Video Quality Platform team reporting on global video analytics for internal use

Tech Stack: Golang, Python, SQL

- Built asynchronous Spark, Kafka and Clickhouse query engine querying **25 petabytes of video analytics data** in under 60 seconds, giving 90+ SREs near-real-time quality of service dashboards
- Worked with the US / EU technical operators to convert requirements into **Chinese documentation** for CN dev teams enabling global teams to access video quality dashboards
- Designed and built analytics to track **SLO, query response times and other metrics** for the Video Quality Platform across 5 regions
- Upgraded the platform codebase from 0% unit test code coverage to **42% in 4 months** and built **2 MCP** server endpoints for LLM access increasing AI adoption across the service

Trainline Engineering Team Lead

📅 May 2023 - June 2024 📍 London, United Kingdom

Team lead for the Ecommerce team handling all payments for train tickets across Europe

Tech Stack: C#, SQL

- Engineering team lead for 10 engineers handling **£4.3 billion** in sales annually
- Drove feature specifications, timelines & company wide negotiations for migration projects worth **£500,000** from SQL to AWS DynamoDB and OpenSearch
- Introduced new Jira service portal to handle and triage 20+ weekly technical requests increasing productivity by **10%** and accelerated team velocity by **20%** by splitting team into squads
- Planned engineering team OKRs and got **2** junior engineers promoted to senior engineers within 1 year

Certifications

2016 - Microsoft 70-461 : Querying SQL Server • 2021 - Udemy: Kotlin for Java Developers

Open Source

Contributor: GraphQL-Markdown
Contributor: DJI Android Mobile SDK

Personal Project

Built and deployed a token on Solana to learn on-chain program development

Experience

Atlassian Senior Software Engineer

 May 2020 – May 2023  Sydney, Australia

Founding engineer for the developer platform portal 'Compass' which is used to manage software services in one tool

Tech Stack: Kotlin, Java, ReactJS

- Brought daily active users in 'Compass' from 0 to **8000** over 9 months, driving company adoption metrics
- Organised **10+** customer interviews to gather functional requirements for the 'activity feed feature' in Compass
- Successfully shipped the 'activity feed' feature to show software lifecycle for a service to **400** daily active users using GraphQL, PostgreSQL and ReactJS
- Lead developer for public GraphQL API and **wrote GraphQL linter** to automate GraphQL validation across the company
- Built Compass and Jira 'incident issue type' sync by efficiently parsing and filtering **1 million requests** per minute using minimal memory footprint
- Interim engineering manager for **7 months**, mentored 8 developers & conducted over 20 technical interviews for team hires

StepStone Group Senior Software Engineer

 May 2018 – April 2020  London, United Kingdom

Tech Stack: C#, NodeJS, SQL, Terraform

- Built a producer & consumer microservice emailing system capable of sending **2 million** emails per day
- Designed and built 4 high volume microservice systems (upwards of **1 million requests**) per day that handled email subscriptions, job searching and event tracking
- Architected and built a ReactJS & NextJS application that came in 3 flavors : iFrame, embedded widget and a fully hosted website for 3rd party clients to integrate with and integrated these with **6 clients**
- Travelled to overseas offices to train offsite teams in writing high performing C# code and AWS queuing technologies such as SES and SNS

Leap Legal Software Software Engineer

 Nov 2016 – May 2018  Sydney, Australia

Tech Stack: C#, SQL

- Helped migrate backend search services to OpenSearch for **20+ law firms**

PwC Software Engineer

 Jan 2013 – July 2016  Sydney, Australia

Tech Stack: C#, SQL

- Migrated **7 databases across 3 countries** to new data models for internal enterprise software division

Publications

[1] M. D. A. Fotouhi and M. Hassan. Dronecells: Improving 5g spectral efficiency using drone-mounted flying base stations. *IEEE Transactions on Mobile Computing*, 2018.